For example,

(i) Consider the experiment of tossing a coin

If X (Head) =1, X (Tail) = 0

Then X takes the values either 0 or 1

This is a discrete random variable.

NOTE

Example 9.1, 9.2, and 9.3 are the random variables taking finite number of values, therefore they are discrete.

(ii) Consider the experiment of tossing a coin till head appears.

Let random variable X denote the number of trials needed to get a head. The values taken by it will be 1, 2, 3, ..

It is discrete random variable taking countable infinite values.

9.2.2 Continuous random variable:

A random variable X is said to be continuous, if it takes values in an interval or union of disjoint intervals. (A rigorous definition is beyond the scope of the book).

For example,

(i) If X is defined as the height of students in a school ranging between 120 cms and 180 cms, Then the random variable X is $\{x/120\ cms < x < 180\ \text{cms}\}$ is a continuous random variable.

(ii) Let the maximum life of electric bulbs is 1500 hrs. Life time of the electric bulb is the continuous random variables and it is written as $X = \{x/0 \leq x \leq 1500\}$

9.3 Probability mass function and probability density function

A probability function is associated with each value of the random variable. This function is used to compute probabilities for events associated with the random variables. The probability function defined for a discrete random variable is called probability mass function. The probability function associated with continuous random variable is called probability density function.

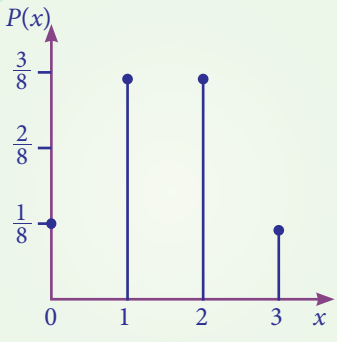


Fig.9.3